

JKBOSE 2026

CLASS 11TH

XUV (For you).

Bhaichara
ON
TOP.

EQUILIBRIUM

LECTURE-02 CHEMISTRY

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Topics To Covered

Chemical Equilibrium

1

Dynamic Nature of Chemical Equilibrium

2

Law of Mass Action

} Important.

REVISION



Equilibrium

Physical Equilibrium

P, T + C
does not change.

Chemical Equilibrium ✓

- a) $S \rightleftharpoons L$ $\xrightarrow{\text{H}_2\text{O}(s) \rightleftharpoons \text{H}_2\text{O}(l)}$
- b) $L \rightleftharpoons G$ (Manometer).
- c) $S \rightleftharpoons V$ (Iodine).
- d) $S \rightleftharpoons \text{Solution}$ (NaCl + H_2O).
- e) $G \rightleftharpoons \text{Solution}$ (Carbonated drinks).

#CHEMICAL EQUILIBRIUM.

In case of reversible chemical reactions, if the rate of forward reaction becomes equal to the rate of backward reaction, then this state of process is called chemical equilibrium.

For example

i) Formation of NH_3



ii) Formation of HI



Dynamic Nature of Chemical Equilibrium

☆☆☆



In case of any reaction which is in chemical equilibrium, pressure, concentration and temperature do not change, it seems that reaction has stopped. but this is not the case. The reaction is occurring continuously i.e. equilibrium is dynamic in nature not static.

Let us take the example of formation of ammonia.



When reaction is in equilibrium the concentration of reactants and products becomes constant. Now if we replace H_2 with its isotope D_2 , then at equilibrium

$\text{N}_2 + 3\text{D}_2 \rightleftharpoons 2\text{ND}_3$ again the concentration observed was constant at equilibrium.

Now, if we add the mixture of H_2 & D_2 in reaction. we found that concentration of all reactants and products were constant at equilibrium. and in reactants N_2 , H_2 , D_2 & HD was found & in products NH_3 , ND_3 , NHD_2 & NH_2D was found. This proves that equilibrium is not static but dynamic in nature.



Characteristics – of Equilibrium

Catalyst increases
the rate of reaction.

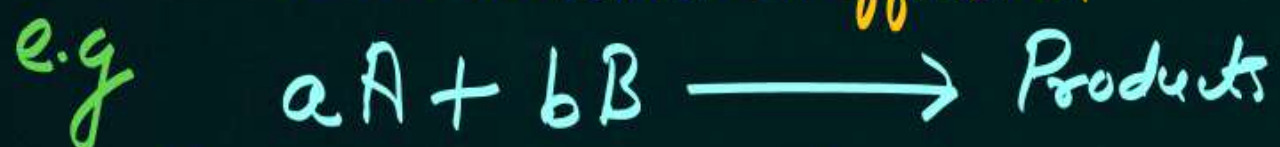


The important characteristics of chemical equilibrium are as follows.

- (i) In chemical equilibrium, at a given temperature, the measurable properties like pressure, concentration, density or colour remain constant.
- (ii) Equilibrium can be attained from either side.
- (iii) Equilibrium remains unaffected if the reaction occurs in the presence of a catalyst. However, equilibrium can be attained in a lesser time by the use of a catalyst.
- (iv) Equilibrium is dynamic in nature (i.e. the reaction does not stop but continues in both the directions with same speed).
- (v) The chemical equilibrium will continue indefinitely if the conditions such as temperature and concentration are not changed.

Law of Mass Action

This law was given by Guldberg & Waage.
It states that, "in a balanced chemical equation, rate of any reaction is directly proportional to the active masses of reactants raise to power their stoichiometric coefficients".



According to law of mass action

$$\text{Rate} \propto [A]^a [B]^b$$

$$\text{Rate} = k[A]^a [B]^b$$

here k = Rate constant.

Consider a equilibrium equation



According to law mass action

Rate of forward Reaction (R_f)



$$R_f = k_f [A]^a [B]^b$$

• Rate of backward Reaction (R_b)



$$R_b = k_b [C]^c [D]^d$$

We know that at equilibrium

$$R_f = R_b$$



$$K_f [A]^a [B]^b = K_b [C]^c [D]^d$$

$$\frac{K_f}{K_b} = \frac{[C]^c [D]^d}{[A]^a [B]^b}$$

$$K_{eq} = \frac{[C]^c [D]^d}{[A]^a [B]^b} \quad \left\{ \therefore \frac{K_f}{K_b} = K_{eq} \right\}$$

We can say that, ratio of multiplication of concentration of products raise to power their stoichiometric coefficient to the multiplication of concentration of reactants raise to power their stoichiometric coefficient is always constant, this called law of chemical equilibrium. Constant is known as equilibrium constant for equilibrium reaction.

QUESTION



PCl_5 , PCl_3 , and Cl_2 are at equilibrium at 500 K in a closed container and their concentrations are $0.8 \times 10^{-3} \text{ mol L}^{-1}$, $1.2 \times 10^{-3} \text{ mol L}^{-1}$ and $1.2 \times 10^{-3} \text{ mol L}^{-1}$, respectively. The value of K_c for the reaction:

$\text{PCl}_5(\text{g}) \rightleftharpoons \text{PCl}_3(\text{g}) + \text{Cl}_2(\text{g})$ will be

1 $1.8 \times 10^{-3} \text{ mol L}^{-1}$

2 1.8×10^{-3} X

3 $1.8 \times 10^{-3} \text{ mol}^1 \text{ L}$ X

4 0.55×10^4 X

$$\begin{aligned} K_{eq} &= \frac{[\text{PCl}_3][\text{Cl}_2]}{[\text{PCl}_5]} \\ &= \frac{(1.2 \times 10^{-3}) (1.2 \times 10^{-3})}{0.8 \times 10^{-3}} \\ &= 1.8 \times 10^{-3} \text{ mol L}^{-1} \end{aligned}$$

$$\begin{aligned} [\text{PCl}_3] &= 1.2 \times 10^{-3} \\ [\text{PCl}_5] &= 0.8 \times 10^{-3} \\ [\text{Cl}_2] &= 1.2 \times 10^{-3} \end{aligned}$$

QUESTION



False

Which of the following is not a general characteristic of equilibria involving physical processes?

- 1 Equilibrium is possible only in a closed system at a given temperature T
- 2 All measurable properties of the system remain constant T
- 3 All the physical processes stop at equilibrium F
- 4 The opposing processes occur at the same rate and there is dynamic but stable condition T

QUESTION

What is the expression of equilibrium constant for synthesis of NH₃.

1 $\frac{[\text{NH}_3]^2}{[\text{N}_2][\text{H}_2]^3}$

2 $\frac{[\text{NH}_3]^2}{[\text{N}_2]}$

3 $\frac{[\text{N}_2]^2}{[\text{H}_2]}$

4 $\frac{[\text{H}_2]^3}{[\text{NH}_3]}$



$$K_{eq} = \frac{\text{Product}}{\text{Reactant}} = \frac{[\text{NH}_3]^2}{[\text{N}_2][\text{H}_2]^3}$$

[?]

QUESTION



What do you mean by dynamic nature of Equilibrium? Explain.

(2025)

PYQ

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REVISION KARNA ACCHE Se.

Thank # XUV
You